# Year 13 Further Maths - Hyperbolic functions

Similar questions - Modelling a hanging rope

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

**1** Two vertical poles *OA* and *BC* stand on horizontal ground, with $O = (0,0)$, $A = (0,5)$ and $B = (7,0)$. A rope is fixed at $A$ and $C$ and just touches the ground at $D = (3,0)$ (units are metres). Jofra models the rope by $y = k\cosh(ax-b)-1$ where $k, a, b > 0$.

**(a)** Determine the value of $k$. **[2]**

**(b)** Find the exact values of $a$ and $b$, in logarithmic form. **[5]**

**(c)** Hence find the height of the pole *BC* to 3 significant figures. **[2]**

**(d)** Holly models the rope by a parabola with minimum at $D$ passing through $A$. Find the height of *BC* predicted by Holly's model and the difference between the two models. **[4]**

## Answers



(a) The minimum of $y = k\cosh(ax-b)-1$ is $k-1$, which is 0 at $D$, so $k=1$
(b) The minimum is at $x=3$ so $3a-b=0$, $b=3a$. $A$: $\cosh(-b)-1=5$, so $\cosh b=6$ and $b=\cosh^{-1}6=\ln(6+\sqrt{35})$.
$a=\frac{1}{3}\ln(6+\sqrt{35})$
(c) At $x=7$: $y=\cosh(7a-b)-1=\cosh(4a)-1=\cosh\left(\frac{4}{3}b\right)-1=12.63$ m, i.e. 12.6 m
(d) $y=\frac{5}{9}(x-3)^2$ passes through $A$ ($\frac{5}{9}\cdot 9=5$). At $x=7$: $\frac{5}{9}\cdot 16=\frac{80}{9}=8.89$ m. The difference is about 3.74 m